

CSE/PC/B/S/314 Computer Networks Lab

CO3: Design and implement medium access control mechanisms within a simulated network environment using IEEE 802 standards.

Assignment 3: Implement CSMA techniques with collision detection.

Code demonstration due on: 21-25 September 2026 (in your respective lab classes)

Report submission due on: next lab classes

Softcopy of the report to be uploaded in the drive.

Language: You can write the program in any high level language like C, C++, Java, Python etc.

Please note that you may need to use these schemes separately for other applications (assignments).

Use the same sender-receiver design as previous assignments.

In this assignment, you have to emulate the shared channel (maintaining its state as IDLE, BUSY, or COLLISION), while multiple independent client processes emulate network stations that sense the channel and attempt transmission according to a given CSMA strategy (Non-Persistent, 1-Persistent, p-Persistent, CSMA/CD). Write methods/functions to emulate carrier sensing, collision detection, and backoff behavior. Analyze how the number of contending stations affects the number of collisions, average transmission delay, and overall channel throughput.

Test the above schemes for the following cases (not limited to).

- i) For p-persistent CSMA, plot number of collisions, transmission delay, and throughput as functions of p (fixed N).
- ii) For all schemes (Non-Persistent, 1-Persistent, p-Persistent at best p, CSMA/CD), plot the same three metrics as functions of the number of contending stations N, and compare their efficiency.

State your observations on the impact of performance of CSMA techniques in different scenarios.